

For the second and third cases we recall Petersen's Theorem [8], which states that if  $k$  is even, every  $k$ -regular graph is the union of  $k/2$  2-factors. Recalling the definition of the chromatic index, if a  $k$ -regular graph has  $\chi'(G) = k$  then it is the union of  $k$  perfect matchings, and, in particular, the union of any two perfect matchings is a 2-factor. ■

## 2.4 Small graphs

We begin this subsection with the following proposition, which, although asymptotically inferior to the result of Theorem 1.1, is more useful for small graphs.

**Proposition 2.5** *If  $G$  is a connected graph with minimum degree  $\delta$  then  $rc(G) \leq n - \delta$ .*

**Proof:** The proof is trivial in the case  $\delta = 1$ . Fixing  $\delta$ , we prove the proposition by induction on  $n$  where the base case  $n = \delta + 1$  is trivial since cliques have rainbow connection number 1. So, we assume  $n > \delta + 1$ .

Let  $K$  be a maximal clique of  $G$  consisting only of vertices whose degree is  $\delta$ . Since there is at least one vertex with degree  $\delta$  and since  $G$  is connected we have  $1 \leq k = |K| \leq \delta$ .

Consider the graph  $G'$  obtained from  $G$  by deleting the vertices of  $K$ . Suppose the connected components of  $G'$  are  $G_1, \dots, G_t$  where  $G_i$  has  $n_i$  vertices and minimum degree  $\delta_i$  for  $i = 1, \dots, t$ . Let  $K_i \subset K$  be the vertices of  $K$  with a neighbor in  $G_i$ , and assume that  $|K_1| \geq |K_i|$  for  $i = 2, \dots, t$  (notice that it may be that  $t = 1$  and  $G'$  is connected). Consider first the case where  $K_1 = K$ . By the induction hypothesis,  $rc(G_i) \leq n_i - \delta_i$ . Clearly, we may give the edges of  $K$  and the edges from  $K$  to  $G_1$  the same color. Hence,

$$rc(G) \leq t + \sum_{i=1}^t (n_i - \delta_i) = t + n - k - \sum_{i=1}^t \delta_i.$$

By the maximality of  $K$ , each vertex of  $G_i$  has degree at least  $\delta - k + 1$  in  $G_i$ . Hence,  $\delta_i \geq \delta - k + 1$ , and therefore

$$rc(G) \leq t + n - k - t\delta + tk - t \leq n - t\delta + (t - 1)k \leq n - \delta.$$

Now assume that  $K_1 \subsetneq K$  but that  $|K_1| = k_1 > 1$ . By contracting  $K_1$  to a single vertex  $v$ , we obtain a contraction  $G^*$  of  $G$  with  $n - k_1 + 1$  vertices and minimum degree  $\delta - k_1 + 1$ , so by induction  $rc(G^*) \leq n - \delta$ . Now, going back to  $G$ , any edge with both endpoints not in  $K_1$  receives the same color it had in  $G^*$ . Any edge with one endpoint in  $K_1$  receives the color of the edge of  $G^*$  from  $v$  to that other endpoint. Any edge with both endpoints in  $K_1$  receives the color of an edge of  $G^*$  from  $v$  to another vertex in  $K \setminus K_1$ . The resulting coloring makes  $G$  rainbow connected and  $rc(G) \leq n - \delta$ .

Finally, if  $k_1 = 1$  (and since  $K_1 \subsetneq K$  we have  $k \geq 2$ ), contract all of  $K$  into a single vertex  $v$  and notice that the contracted graph  $G^*$  also has minimum degree  $\delta$ , and  $n - k + 1$  vertices. Hence, by induction,  $rc(G^*) \leq n - k - \delta + 1$ . Going back to  $G$  and coloring the edges of the clique  $K$  with another new color, we obtain  $rc(G) \leq n - k - \delta + 2 \leq n - \delta$ . ■

**Corollary 2.6** *If  $G$  is a connected graph with  $n$  vertices and average degree  $d \geq 2$  then  $rc(G) \leq n - d/2 - 1$ .*

**Proof:** It is well known (see, e.g., [2]) that a graph  $G$  with average degree  $d \geq 2$  has a connected subgraph  $H$  with minimum degree at least  $d/2 + 1$ . By proposition 2.5,  $rc(H) \leq |H| - d/2 - 1$ . By Lemma 2.1,  $rc(G) \leq n - |H| + rc(H) \leq n - d/2 - 1$ . ■

## 3 Ensuring 2-rainbow-connection

### 3.1 Dense graphs

**Proof of Theorem 1.5:** Given a graph  $G$  with  $n$  vertices and  $\delta(G) \geq n/2 + \log n$ , we randomly color the edges with two colors, red and blue. We show that with positive probability, such a random coloring makes  $G$  rainbow connected. Consider two non-adjacent vertices  $x, y$ . The minimum degree requirement forces  $x$  and  $y$  to have more than  $2 \log n$  common neighbors. Now, for each such common neighbor  $z$ , the probability that the path  $x, z, y$  is not a rainbow path is precisely  $1/2$ . Since the paths corresponding to distinct common neighbors are edge-disjoint, the probability that *all* these paths are not rainbow is less than  $(1/2)^{2 \log n} = 1/n^2$ . Since there are less than  $\binom{n}{2}$  pairs  $x, y$  to consider, it follows from the union bound that with positive probability, each pair of non-adjacent vertices are connected by a rainbow path. ■

A bipartite graph which is not complete has diameter at least 3. A proof similar to the proof of Theorem 1.5 gives the following result.

**Theorem 3.1** *Let  $c = 1/\log(9/7)$ . If  $G$  is a non-complete bipartite graph with  $n$  vertices and any two vertices in the same vertex class have at least  $2c \log n$  common neighbors in the other vertex class, then  $rc(G) = 3$ .*

**Proof:** The only modification needed is that now the edges are colored with three colors. For two vertices with distance 2 the probability that a common neighbor is the center of a non-rainbow path is only  $1/3$ . For two vertices with distance 3 (clearly the diameter is at most 3), say,  $x, y$ , fixing a neighbor  $z$  of  $x$ , and the color  $c$  of  $(x, z)$ , the probability that a common neighbor of  $y$  and  $z$  is the center of a non-rainbow path, or a path that contains  $c$ , is only  $7/9$ . Since  $(7/9)^{2c \log n} = 1/n^2$  the result follows as before. ■

### 3.2 Random graphs

**Proof of Theorem 1.6:** For the first part of the theorem, we need to prove that for a sufficiently large constant  $C$ , the random graph  $G(n, p)$  with  $p = C\sqrt{\log n/n}$  almost surely has  $rc(G(n, p)) = 2$ . By the proof of theorem 1.5, it suffices to show that almost surely any two vertices of  $G(n, p)$  have at least  $2 \log n$  common neighbors. By the union bound,

it suffices to prove that a fixed pair of vertices does not have at least  $2 \log n$  common neighbors with probability  $o(1/n^2)$ . Fixing a pair  $x, y$ , the probability of the event that  $z$  is a common neighbor is  $C^2 \log n/n$ . Thus, the expected number of common neighbors is  $\frac{n-2}{n}(C^2 \log n) > 0.5c^2 \log n$ , and this is the sum of  $n - 2$  indicator random variables of *independent* events. By the standard Chernoff estimates (cf. [1] Theorem A.1.13), the probability that the number of common neighbors is less than half of its expectation (in this case, less than  $0.25c^2 \log n$ ) is  $o(1/n^2)$  for  $C$  sufficiently large.

For the other direction, it suffices to prove that for a sufficiently small constant  $c$ , the random graph  $G(n, p)$  with  $p = c\sqrt{\log n/n}$  almost surely does not have diameter two. Fix a set  $X$  of  $n^{1/5}$  vertices (we may and will assume that  $n^{1/5}$  is an even integer), and let  $Y$  be the remaining  $n - n^{1/5}$  vertices. Let  $\mathcal{A}$  be the event that  $X$  induces an independent set, and let  $\mathcal{B}$  be the event that there exists a pair of vertices of  $X$  with no common neighbor in  $Y$ . Clearly, if both  $\mathcal{A}$  and  $\mathcal{B}$  occur then the diameter is at least 3. Hence, it suffices to prove (separately) that for a small  $c$ ,  $\mathcal{A}$  occurs with probability approaching 1 and  $\mathcal{B}$  occurs with probability approaching 1.

We start with  $\mathcal{A}$  first. For  $c$  sufficiently small we indeed have:

$$\Pr[\mathcal{A}] = (1 - p)^{\binom{n^{1/5}}{2}} = (1 - c\sqrt{\log n/n})^{\binom{n^{1/5}}{2}} = 1 - o_n(1).$$

To estimate  $\mathcal{B}$ , let us first partition  $X$  into  $n^{1/5}/2$  pairs, arbitrarily. For a pair  $x, y$ , the probability that  $x, y$  have a common neighbor in  $Y$  is precisely

$$1 - \left(1 - \frac{c^2 \log n}{n}\right)^{n - n^{1/5}}.$$

Thus, for  $c$  sufficiently small, the probability that all  $|X|/2$  pairs have a common neighbor is

$$\left(1 - \left(1 - \frac{c^2 \log n}{n}\right)^{n - n^{1/5}}\right)^{n^{1/5}/2} = o_n(1).$$

We use here the fact that the event that a pair  $x, y$  has a common neighbor in  $Y$  is *independent* of all the events that other pairs have common neighbors in  $Y$ . Since the last inequality is  $o_n(1)$  we have that with probability  $1 - o_n(1)$  at least one of the pairs does not have a common neighbor in  $Y$ , and, in particular,  $\mathcal{B}$  holds. ■

## 4 Concluding remarks and open problems

- The computational status of rainbow connection is not yet determined. In fact, we do not even know whether deciding if  $rc(G) = 2$  is polynomial or is NP-complete. Recall that it is straightforward to determine the diameter of a graph in polynomial time. The main difficulty, however, lies in the fact that even if we know that  $\text{diam}(G) = 2$  (and, in particular, know that any two non-adjacent vertices have at least one common neighbor), we still cannot even use this fact to bound the rainbow connection number. We do conjecture the following: